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United States Patent	12397458
Kind Code	B2
Date of Patent	August 26, 2025
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Razor blade

Abstract

A razor blade including a substrate with a cutting edge portion ending in a sharpened tip. The substrate having a thickness of between 1.55 and 1.97 micrometers measured at a distance of five micrometers from the tip, a thickness of between 4.6 and 6.34 micrometers measured at a distance of twenty micrometers from the tip, and a thickness of between 19.8 and 27.12 micrometers measured at a distance of one hundred micrometers from the tip.

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Appl. No.: 17/551841

Filed: December 15, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220105648 A1	Apr. 07, 2022

Related U.S. Application Data

continuation parent-doc US 15535984 US 11230024 WO PCT/EP2014/079091 20141222 child-doc US 17551841

Publication Classification

Int. Cl.: B26B21/56 (20060101); B26B21/58 (20060101)

U.S. Cl.:

CPC **B26B21/56** (20130101); B26B21/58 (20130101)

Field of Classification Search

CPC: B26B (21/54); B26B (21/56); B26B (21/565); B26B (21/60)

USPC: 30/47-51; 30/346.5; 30/346.53-3; 30/346.58; 30/350; 30/357; 76/104.1; 76/DIG.8

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3761372	12/1972	Sastri	N/A	N/A
3835537	12/1973	Sastri	30/346.55	B26B 21/54
3911579	12/1974	Lane et al.	N/A	N/A
4720918	12/1987	Curry et al.	N/A	N/A
4916817	12/1989	Atwater	30/346.55	B24B 3/48
5032243	12/1990	Bache	156/345.39	C23F 4/00
5048191	12/1990	Hahn	N/A	N/A
5056227	12/1990	Kramer	76/DIG.8	B26B 21/54
5088195	12/1991	Lazarshik	30/346.55	B26B 21/56
5088202	12/1991	Boland	76/DIG.8	B26B 21/58
5129289	12/1991	Boland et al.	N/A	N/A
5142785	12/1991	Grewal	76/DIG.8	B26B 21/60
5275672	12/1993	Althaus et al.	N/A	N/A
5497550	12/1995	Trotta	30/346.55	B26B 21/60
5669144	12/1996	Hahn et al.	N/A	N/A
5983756	12/1998	Orloff	76/DIG.8	B26B 21/20
6105261	12/1999	Ecer	76/DIG.8	B26B 21/58
6289593	12/2000	Decker	30/346.54	B26B 21/60
6327784	12/2000	Altena	30/346.54	B26B 21/60
6330750	12/2000	Meckel	N/A	N/A
6468642	12/2001	Bray	428/408	B26B 21/60
6684513	12/2003	Clipstone et al.	N/A	N/A
7060367	12/2005	Yamada et al.	N/A	N/A
7140113	12/2005	King et al.	N/A	N/A
7587829	12/2008	King	30/346.54	B26B 21/58
7758707	12/2009	Millward	N/A	N/A
8011104	12/2010	Masek et al.	N/A	N/A
8322253	12/2011	Howells	N/A	N/A
8409462	12/2012	Keenan	216/101	A61B 17/32
8499462	12/2012	Masek et al.	N/A	N/A
8769833	12/2013	Culf	N/A	N/A
9079321	12/2014	Claus et al.	N/A	N/A
9248579	12/2015	DePuydt et al.	N/A	N/A
9751230	12/2016	Skrobis	N/A	B26B 21/60
9902013	12/2017	Uemura et al.	N/A	N/A

10549438	12/2019	Skrobis et al.	N/A	N/A
11230024	12/2021	Papatriantafyllou et al.	N/A	B26B 21/56
11654588	12/2022	Nisby	30/346.54	B26B 21/56
11660770	12/2022	Lee	30/346.5	B26B 21/60
2003/0208912	12/2002	Park et al.	N/A	N/A
2004/0121159	12/2003	Cloud	30/62	B26B 21/60
2004/0177516	12/2003	Teeuw et al.	N/A	N/A
2005/0160878	12/2004	Wort et al.	N/A	N/A
2006/0201001	12/2005	Teeuw et al.	N/A	N/A
2006/0265885	12/2005	Clipstone	30/346.54	B26B 21/60
2006/0277767	12/2005	Sun et al.	N/A	N/A
2007/0186424	12/2006	Becker	30/294	B26B 21/60
2007/0283578	12/2006	Newman	30/346.54	B26B 21/56
2008/0086888	12/2007	Scheinfeld	30/346.53	B26B 21/60
2008/0178476	12/2007	Luttgens	N/A	N/A
2008/0190758	12/2007	Papachristos	204/192.11	B26B 21/60
2009/0194203	12/2008	Hirai	148/226	B26B 21/60
2010/0011590	12/2009	DePuydt et al.	N/A	N/A
2010/0024222	12/2009	Akari	30/346.54	B26B 21/60
2010/0178493	12/2009	Wang	264/510	B26B 21/60
2010/0178515	12/2009	Wang	428/421	B26B 21/60
2010/0287781	12/2009	Skrobis	30/346.54	B26B 21/60
2010/0288097	12/2009	Uemura	83/651	B26B 21/54
2010/0299931	12/2009	Marchev et al.	N/A	N/A
2011/0010950	12/2010	Madeira	30/346.54	B26B 21/60
2011/0209988	12/2010	Madeira	204/192.16	B26B 21/60
2012/0276826	12/2011	Floter	83/651	B26B 21/60
2012/0311865	12/2011	Hamilton	30/346.5	B26B 21/60
2013/0014395	12/2012	Patel et al.	N/A	N/A
2013/0014396	12/2012	Skrobis et al.	N/A	N/A
2013/0031794	12/2012	Duff, Jr.	30/346.53	B26B 21/60
2015/0328789	12/2014	Skrobis et al.	N/A	N/A
2016/0016322	12/2015	Islam et al.	N/A	N/A
2018/0043561	12/2017	Nisby et al.	N/A	N/A
2018/0215056	12/2017	Logothetidis et al.	N/A	N/A
2020/0307006	12/2019	Kontokostas et al.	N/A	N/A
2020/0316802	12/2019	Kontokostas et al.	N/A	N/A
2021/0094199	12/2020	Lee et al.	N/A	N/A
2023/0311352	12/2022	Lee	30/346.5	B26B 21/56
2023/0311354	12/2022	Claus	30/346.53	B26B 21/60
2023/0347537	12/2022	Maziarz	N/A	B26B 21/56
2023/0364818	12/2022	Park	N/A	B26B 21/56
2023/0373119	12/2022	Skrobis	N/A	B26B 21/60
2023/0373120	12/2022	Skrobis	N/A	B26B 21/60
2023/0373121	12/2022	Skrobis	N/A	B26B 21/60
2023/0373122	12/2022	Skrobis	N/A	B26B 21/60
2024/0051167	12/2023	Duff, Jr.	N/A	B26B 21/60
2024/0051168	12/2023	Duff, Jr.	N/A	B26B 21/60
2024/0051169	12/2023	Duff, Jr.	N/A	B26B 21/60
2024/0051170	12/2023	Duff, Jr.	N/A	B26B 21/60

2024/0131740	12/2023	Gluche	N/A	B26B 21/56
2024/0246251	12/2023	Jung	N/A	B26B 21/56
2024/0300126	12/2023	Park	N/A	B26B 21/56
2024/0308094	12/2023	Lee	N/A	B26B 21/56
2024/0316810	12/2023	Lee	N/A	B26B 21/60
2024/0383163	12/2023	Pandis	N/A	B26B 21/60
2025/0135670	12/2024	Hanson	N/A	B26B 21/60
2025/0153381	12/2024	Jung	N/A	B26B 21/60
2025/0153382	12/2024	Jung	N/A	B26B 21/56

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
29718352	12/1996	DE	N/A
1 465 697	12/1976	EP	N/A
0 126 128	12/1986	EP	N/A
0 363 648	12/1992	EP	N/A
0 612 280	12/1997	EP	N/A
0 255 834	12/1997	EP	N/A
0 589 756	12/1997	EP	N/A
0 884 142	12/1997	EP	N/A
0 582 676	12/1999	EP	N/A
1 259 361	12/2001	EP	N/A
0 757 615	12/2003	EP	N/A
1 397 234	12/2005	EP	N/A
1 601 507	12/2009	EP	N/A
2 323 819	12/2011	EP	N/A
2 326 470	12/2011	EP	N/A
2 495 080	12/2013	EP	N/A
03/006218	12/2002	WO	N/A
2006/079361	12/2005	WO	N/A
2007/095120	12/2006	WO	N/A
2008/065949	12/2007	WO	N/A
2009/143130	12/2008	WO	N/A
2010/008981	12/2009	WO	N/A
2013/010049	12/2012	WO	N/A
2013/010072	12/2012	WO	N/A
2015/179217	12/2014	WO	N/A

OTHER PUBLICATIONS

International Search Report of PCT/EP2014/079091, dated Aug. 18, 2015. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 15/535,984, filed Jun. 14, 2017, issued as U.S. Pat. No.

11,230,024 on Jan. 25, 2022, which is a National Stage application of International Application No. PCT/EP2014/079091, filed on Dec. 22, 2014, each of which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND OF THE INVENTION

Field of the Disclosure

(1) The disclosure relates to razors and more particularly to razor blades wherein the cutting area of the razor blade is profiled.

Description of the Related Art

(2) The shape of a razor blade plays an important role in the quality of a shave. The blade typically has a continuously tapering shape converging toward an ultimate tip. The portion of the blade which is closest to the ultimate tip is called the tip edge.

(3) If the tip edge is robust, it will enable less wear and a longer service life, but it would result in larger cutting forces, which adversely affect shaving comfort. A thin tip edge profile leads to less cutting forces but also to an increase in risk of breakage or damage, and a shorter service life. Therefore, an optimal trade-off between the cutting forces, the shaving comfort and the service life of the cutting edge of a razor blade is desired.

(4) To achieve this optimal trade-off, the cutting edge of the razor blade is shaped using a grinding process.

(5) Historically, there has been numerous technologies related to the geometry of some specific parts of the blade. A typical example includes a known technology that focuses on the geometry of the ultimate tip of the blade. The technology precisely defines the geometry of the tip up to 8000 Angstroms, that is 0.8 micrometers from the tip. This geometry mostly relates to the entry of the blade inside the hair to be cut (the diameter of which is generally of the order of 100 micrometers).

(6) Other technologies focus on an overall view of the whole blade geometry. One example of uses numerical data and a specific angle to define the blade tip.

(7) Other examples of technologies focus on blade geometries that are related to thinning the blade within micrometers of the tip, in relation to a specific angle furthest away from the tip.

(8) Still other examples of blade technologies focus on improving the tip shape by using a hyperbolic equation or constant facet convergence towards the tip of the blade to define the shape of the tip, with respect to a distance within micrometers from the tip.

SUMMARY OF THE INVENTION

(9) It is an object of the disclosure to provide a razor blade, suitable for a shaving head of a shaver, wherein the wear of the razor blade may be reduced and the service life may be further extended, while the cutting forces may at least be equally small and the shaving comfort may at least be equally high.

(10) According to an aspect, a razor blade substrate may include a symmetrical tapering cutting edge ending in a sharpened tip. The substrate may have a continuously tapering geometry toward the tip with a thickness of between 1.55 and 1.97 micrometers measured at a distance of five micrometers from the tip, a thickness of between 4.60 and 6.34 micrometers measured at a distance of twenty micrometers from the tip, a thickness of between 19.80 and 27.12 measured at a distance of hundred micrometers from the tip. Unless explicitly stated otherwise, all blade edge measurement data provided herein are obtained through confocal microscopy measurements.

(11) It has been found that the definition of the geometry of the profile, according to the above-specified key points, may be essential to define a properly supported thin edge tip, which would in turn provide an optimal trade-off between shaving performance, in terms of comfort, since such a profile results in low cutting forces and adequate service life, due to the resulted geometry and the thickness beyond the 20 μm area from the ultimate tip.

(12) According to an aspect, the substrate may have a thickness of between 6.50 and 8.94 micrometers measured at a distance of thirty micrometers from the tip.

- (13) According to an aspect, the substrate may have a thickness of between 8.40 and 11.54 micrometers measured at a distance of forty micrometers from the tip.
- (14) According to an aspect, the substrate may have a thickness of between 10.30 and 14.13 micrometers measured at a distance of fifty micrometers from the tip.
- (15) According to an aspect, the substrate may have a thickness of between 29.30 and 40.11 micrometers measured at a distance of hundred fifty micrometers from the tip.
- (16) According to an aspect, the substrate may have a thickness of between 38.80 and 49.74 micrometers measured at a distance of two hundred micrometers from the tip.
- (17) According to an aspect, the substrate may have a thickness of between 48.30 and 59.37 micrometers measured at a distance of two hundred fifty micrometers from the tip.
- (18) According to an aspect, the substrate may have a thickness of between 57.80 and 69.00 micrometers measured at a distance of three hundred micrometers from the tip.
- (19) According to an aspect, the substrate may have a thickness of between 67.30 and 78.62 micrometers measured at a distance of three hundred fifty micrometers from the tip.
- (20) According to an aspect, the substrate of the razor blade may have a thickness of between 1.80 and 1.95 micrometers measured at a distance of five micrometers from the tip.
- (21) According to an aspect, the substrate of the razor blade may have a thickness of between 5.40 and 6.30 micrometers measured at a distance of twenty micrometers from the tip.
- (22) According to an aspect, the substrate of the razor blade may have a thickness of between 7.00 and 8.00 micrometers measured at a distance of thirty micrometers from the tip.
- (23) According to an aspect, the substrate may have a thickness of between 9.20 and 10.70 micrometers measured at a distance of forty micrometers from the tip.
- (24) According to an aspect, the substrate of the razor blade may have a thickness of between 11.20 and 13.10 micrometers measured at a distance of fifty micrometers from the tip.
- (25) According to an aspect, the substrate of the razor blade may have a thickness of between 23.00 and 25.10 micrometers measured at a distance of hundred micrometers from the tip.
- (26) According to an aspect, the substrate of the razor blade may have a thickness of between 32.30 and 37.10 micrometers measured at a distance of hundred fifty micrometers from the tip.
- (27) According to an aspect, the substrate of the razor blade may have a thickness of between 41.00 and 47.30 micrometers measured at a distance of two hundred micrometers from the tip.
- (28) According to an aspect, the substrate of the razor blade may have a thickness of between 51.40 and 56.50 micrometers measured at a distance of two hundred fifty micrometers from the tip.
- (29) According to an aspect, the substrate of the razor blade may have a thickness of between 61.00 and 65.40 micrometers measured at a distance of three hundred micrometers from the tip.
- (30) According to an aspect, the substrate of the razor blade may have a thickness of between 70.40 and 76.10 micrometers measured at a distance of three hundred fifty micrometers from the tip.
- (31) According to an aspect, the thickness of the cutting edge of the substrate may be described with the following mathematical formulas:

$$t=a.Math.(xb) \quad (A)$$

$$t=(c.Math.x)+d \quad (B)$$

wherein, in formulas A and B, a and c are constants from an interval (0, 1), b is a constant from an interval (0.5, 1), d is a constant from an interval (0.5, 20), x refers to a distance from the tip in micrometers and t refers to the thickness of the blade in micrometers, and wherein equation A is applied from the tip to a transition point, and either equation A or equation B elsewhere. According to an aspect, the substrate may contain a stainless steel including in weight: 0.62-0.75% of carbon, 12.7-13.7% of chromium, 0.45-0.75% of manganese, 0.20-0.50% of Silicon, No more than traces of Molybdenum, Balanced iron. Balanced iron can mean that iron comprises the rest of the material by weight.

(32) According to an aspect, the substrate may be covered by a strengthening coating.

(33) According to an aspect, the strengthening coating comprises Titanium and Boron.

- (34) According to an aspect, the substrate may be covered by an interlayer, and the interlayer may be covered by said strengthening layer.
- (35) According to an aspect, the strengthening layer may be covered by a top layer.
- (36) According to an aspect, the top layer may be covered by a polytetrafluoroethylene layer.
- (37) According to some aspects, the thickness range between 50 and 350 μm distance from the tip may be important in order to achieve the desired geometry for shaving comfort and blade durability.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other characteristics and advantages of the disclosure will readily appear from the following description of some of embodiments, provided as non-limitative examples, and of the accompanying drawings.
- (2) In the drawings:
- (3) FIG. 1 is a profile view of the ultimate tip of a razor blade;
- (4) FIG. 2 is a profile view of the cutting edge of a razor blade;
- (5) FIG. 3 is a profile view of a cutting edge of a razor blade covered by coating layers;
- (6) FIG. 4 is a profile view of a cutting edge of a razor blade covered by coating layers;
- (7) FIG. 5 is a schematic view of the confocal microscopy measurement system,
- (8) FIGS. 6 and 7 are schematic views of a grinding machine,
- (9) FIGS. 8a and 8b are cross-sectional views of two aspects of a razor blade.
- (10) On the different Figures, the same reference signs designate like or similar elements.

DETAILED DESCRIPTION

(11) According to aspects of the disclosure, the desired blade profile may be achieved by a grinding process that involves two, three or four grinding stations. FIG. 6 schematically shows a grinding installation 1 having two stations 2a and 2b. The base material may be a continuous strip 3. The continuous strip 3 may be made of the raw material for the razor blade substrate, which may have been previously submitted to a suitable metallurgical treatment. This, for example, the continuous strip 3 may be stainless steel. The disclosure may also be applicable to razor blades with a substrate of carbon steel. According to other aspects, ceramic may be a possible material. These materials may be considered insofar as being suitable for razor blade materials. The continuous strip 3 may be longer than a plurality of razor blades, for example the continuous strip 3 may correspond to approximately 1000 individual razor blades or more. Before grinding, the continuous strip 3 may, generally speaking, be rectangular in cross-section. The height of the continuous strip 3 may be slightly over the height of one finished razor blade, or slightly over the height of two finished razor blades, if grinding is to be performed on both edges. The thickness of the continuous strip 3 may be the maximum thickness of the future razor blades. The continuous strip 3 may include through punches which may enable the continuous strip 3 to be carried along the installation 1 during the grinding process, and/or may be used to facilitate future separation of the individual razor blades from the continuous strip 3.

(12) As the metal strip 3 moves along the grinding stations 2a, 2b, continuous strip 3 may be sequentially subjected to a rough grinding, a semi-finishing and a finishing grinding operation. Depending on the number of stations involved, the rough grinding and semi-finishing operation may be performed separately or in the same station. Thereafter, a finishing grinding operation may be required. The grinding steps may be performed continuously, in that the continuous strip 3 may be moved continuously through the stations without stopping.

(13) When the rough grinding is performed separately, one or two grinding stations may be required. Each grinding station may utilize one or two abrading wheels that may be positioned

parallel with respect to the moving continuous strip **3**. The abrading wheels may have a uniform grit size along their length. The abrading wheels may also be full body or helically grooved along their length. The material of the abrading wheels may use resin-bonded or vitrified diamond, resin-bonded or vitrified CBN (Cubic Boron Nitride), or resin-bonded or vitrified silicon carbide, aluminum oxide grains or a mixture of the above grains.

(14) When rough grinding and semi-finishing operations are performed simultaneously, a single grinding station may be required for these operations. In this case, the station may include two abrading wheels formed into spiral helixes or a sequence of straight discs with a special profile. The rotational axes of these wheels may be parallel or positioned at an angle α_1 with respect to the moving continuous strip **3**. The tilt angle ranges between 0.5 degrees and 2 degrees. The grit size of the wheels may also be uniform or progressively decreasing along their length towards the exit of the strip. The material of the abrading wheels might use resin-bonded or vitrified diamond, resin-bonded or vitrified CBN (Cubic Boron Nitride) or resin-bonded or vitrified silicon carbide, aluminum oxide grains or a mixture of the above grains.

(15) The finishing operation may require a single grinding station with two abrading wheels positioned at an angle with respect to the moving continuous strip **3**. The tilt angle α_2 may be reversed compared to the tilt angle used in the rough grinding operation. The tilted angle may range between 1 degree and 5 degrees. The wheels may form spiral helixes and may be specially profiled. The abrasive material may be single grain or multi-grain material from the aforementioned CBN, silicon carbide, aluminum oxide or Diamond.

(16) The process may be tuned so as to obtain a symmetrical razor blade substrate **10** with a continuously tapering geometry toward the tip, as shown in FIG. 2.

(17) For the measurement of the blade geometry, surface roughness and grinded angle, a confocal microscope may be used. For example, is shown on FIG. 5, the confocal microscope may include a LED light source **21**, a pinhole plate **22**, an objective lens **23** with a piezo drive **24** and a CCD camera **25**. The LED source **21** may be focused through the pinhole plate **22** and the objective lens **23** on to the sample **26** surface, which reflects the light. The reflected light may be reduced by the pinhole of the pinhole plate **22** to that part which may be in focus, and this falls on the CCD camera. The sample **26** shown here may not necessarily represent a razor blade. The razor blade may be used with a side angled with respect to the lens focus axis passing through the lens **23** within the device. The confocal microscope may have a given measurement field of, for example, 200 μm $\times$ 200 μm . A semi-transparent mirror **28** may be used between the pinhole plate **22** and the lens **23** to direct the reflected light toward the CCD **25**. Another pinhole plate **27** may be used for the filtering. However, the semi-transparent mirror **28** may be used between the light source and the pinhole plate **22**, which may enable the use of only one pinhole plate for both the emitted light signal and the reflected light signal.

(18) The piezo-drive **24** may be adapted to move the lens **23** along the light propagation axis, to change the position of the focal point in depth. The focal plane may be changed while keeping the dimensions of the measurement field.

(19) To extend the measurement field (in particular, in order to measure the blade edge further away from the tip), another measurement may be performed at another location, and the data resulting from all measurements may be stitched.

(20) The other side of the blade may then be measured, simply by flipping the blade to the other side.

(21) According to one aspect, a confocal microscope based on the Confocal Multi Pinhole (CMP) technology may be used.

(22) The pinhole plate **22** may have a large number of holes arranged in a special pattern. The movement of the pinhole plate **22** may enable seamless scanning of the entire surface of the sample within the image field and only the light from the focal plane may reach the CCD camera, with the intensity following the confocal curve. Thus, the confocal microscope may be capable of high

resolution in the nanometer range.

(23) Also, other methods may be used to measure the thickness of the razor blade. For example, measuring the cross-section of the blade by a Scanning Electron Microscope (SEM). SEM may be performed on a blade cross-section. Currently, SEM may be limited with regards to providing relevant measurement data because it is compulsory to prepare a cross-section of the razor blade. The preparation of samples to be imaged may be rather difficult, in that very few samples may be imaged, and the results may likely be non-statistically relevant.

(24) According to other aspects, it may also possible to measure the thickness of the blade by an interferometer. For this measurement, white light probes from one of a variety of sources (halogen, LED, xenon, etc.) may be coupled into an optical fiber in the controller unit and transmitted to an optical probe. The emitted light may undergo reflection from the blade and may be collected back into the optical probe, and pass back up the fiber where it is collected into an analysis unit. The modulated signal is may be subjected to a fast Fourier transform to deliver a thickness measurement. However, since this measurement is based on light interference from the surface of the blade, the thickness measured by this method may be adversely affected.

(25) Measurements of the same blade using the same method may be performed at different times by different operators in order to that the method is capable of being repeated. Test have shown that confocal microscopy may offer a much better repeatability and reproducibility than the interferometry method.

(26) Additionally, numerous measurements were carried out with the above-mentioned measurement methods on several blades in order to determine the correct thickness of the cutting edge. The average results of these measurements are depicted in the following Table 1.

(27) TABLE-US-00001 TABLE 1 Comparison of thickness measuring methods Distance from the tip Thickness of the blade [μm] [μm] Interferometer Confocal microscope 4 1.55 1.79 5 1.88 2.16 8 2.84 3.16 16 5.22 5.59 20 6.40 6.74

(28) From the above Table 1, it may be apparent that the results of the interferometry measurement method are different from the results of the confocal microscopy method. Therefore, in view of the better reproducibility of the measurement using confocal microscopy as discussed above, the dimensions are obtained by measurement using the above confocal microscopy method.

(29) According to an aspect, the razor blade, may include a blade substrate **10** which may be sharpened. The blade substrate **10** may have a planar portion **8**, wherein the two opposite sides of the blade may be parallel to each other. Further, the blade substrate **10** may also include a cutting edge portion **11**, shown in cross-section on FIG. **1** and FIG. **2**. The cutting edge portion **11** may be connected to a planar portion **8**, and may have sides **12** and **13** that taper and converge to a substrate tip **14**. The thickness of the cutting edge portion **11** may be measured by a confocal microscope. The shape of the blade may be profiled, meaning that the cross-section of the blade may be roughly identical along the length of the blade.

(30) Razor blades with various geometries may have been manufactured, measured, and tested for shaving performance. Manufacture may include, not only substrate sharpening by grinding, but also coatings as will be described below. For the shaving tests, only the grinding step may be modified in order to generate various substrate geometries, the other process steps may be kept equal.

(31) The tests determined that the thinness of the tip edge may be defined by checking the thickness of control points located 5 and 20 micrometers from the tip. Further, the strength of the edge tip may be defined by checking the thickness of control points located 20 and 100 micrometers from the tip.

(32) Further, the dimensions given here may be average dimensions along the length of the blade. Due to the manufacturing process, a single blade may not have exactly the same profile along its whole length. Hence, each thickness value may be an average value of various data obtained along the length, for example, between 4 and 10 data.

(33) After intense testing, it was determined that suitable shaving effects may be obtained for blades having the following features:

(34) The cutting edge portion **11** of the blade may have a thickness of **T5** between 1.55 and 1.97 micrometers measured at a distance **D5** of five micrometers from the tip.

(35) The cutting edge portion **11** of the blade may have a thickness of **T20** between 4.60 and 6.34 micrometers measured at a distance **D20** of twenty micrometers from the tip.

(36) The cutting edge portion **11** of the blade may have a thickness of **T100** between 19.80 and 27.12 micrometers measured at a distance **D100** of hundred micrometers from the tip.

(37) The above dimensions may be obtained through a dispersion of products manufactured using the same manufacturing process.

(38) The blade may have a smooth profile in between and beyond (both from and away from the tip) these control points. The above-mentioned results may the profiles as detailed in Table 2 (although measured thickness geometry in other check points may not be considered as relevant in terms of qualifying the quality of the product).

(39) TABLE-US-00002 TABLE 2 Suitable blade profile parameters

	Lower	Upper	Distance
thickness	thickness	from tip	limit
limit	[μm]	[μm]	[μm]
5	1.55	1.97	20
4.60	6.34	30	6.50
8.94	40	8.40	11.54
50	10.30	14.13	100
19.80	27.12	150	29.30
40.11	200	38.80	49.74
250	48.30	59.37	300
57.80	69.00	350	67.30
78.62			

(40) According to some aspects, the thickness of the cutting edge portion **11** may have the following configuration of thicknesses. The thickness **T5** may be between 1.80 and 1.95 micrometers measured at a distance **D5** of five micrometers from the tip. The thickness **T20** may be between 5.40 and 6.30 micrometers measured at a distance **D20** twenty micrometers from the tip. The thickness of **T100** may be between 23.00 and 25.10 micrometers measured at a distance **D100** hundred micrometers from the tip.

(41) In such cases, the thickness configuration may be detailed in following Table 3.

(42) TABLE-US-00003 TABLE 3 Suitable blade profile parameters

	Lower	Upper	Distance
thickness	thickness	from tip	limit
limit	[μm]	[μm]	[μm]
5	1.80	1.95	20
5.40	6.30	30	7.00
8.00	40	9.20	10.70
50	11.20	13.10	100
23.00	25.10	150	32.30
37.10	200	41.00	47.30
250	51.40	56.50	300
61.00	65.40	350	70.40
76.10			

(43) According to further aspects, Table 4 may detail examples of thickness configurations.

(44) TABLE-US-00004 TABLE 4 Blade profile parameters

Distance from the tip	Thickness [μm]
[μm]	0
0.00	5
1.81	20
5.49	30
7.60	40
9.56	50
11.50	100
21.50	150
31.50	200
41.50	250
51.50	300
61.50	350
71.50	

(45) The blade thickness increase rate (slope) from the tip up to the transition point may be continuously decreasing, making the blade edge easier to penetrate the hair leading to better comfort. The blade profile after the transition point **100** (for example from 40 μm to 350 μm) may be lying in a specific range of values in order to support a geometrically smooth transition from the first 40 μm to the unground part of the blade. In this region, the thickness increase rate may be less than, or equal to, the increase rate at 40 μm.

(46) The blade edge profile generated by the rough grinding stage, typically covering an area between 50-350 μm from the tip, may determine the material removal rate of the finishing operation. Generally, the finishing grinding stage may be mainly called to smoothen out the excess surface roughness produced by rough grinding along with the final shaping of the blade edge profile. For optimal process efficiency, the material removal rate of finishing grinding wheel may be kept to a minimum but such that the induced surface roughness ranges between 0.005-0.040 μm.

(47) For example, the thickness of the aforementioned blade profile may be described with the following mathematical formulas:

$$t=a.Math.(xb) \quad (A)$$

$$t=(c.Math.x)+d \quad (B)$$

(48) In the above formulas, a and c may be constants from an interval [0, 1], b may also a constant

from an interval [0.5, 1], d may be a constant from an interval [0.5, 20], x may refer to a distance from the tip in micrometers and t may refer to the thickness of the blade in micrometers.

(49) One or more formulas (A) may be applied one after the other to the portion of the blade extending from the tip to a transition point **100**, and one or more formulas (B) may be applied one after the other from the transition point **100** to the unground portion of the blade.

(50) According to some aspects, formula (A) may describe the thickness of the cutting edge portion **11** from 0 to 40 micrometers from the tip, where $a=0.5$ and $b=0.8$ may be constants. Formula (B) may describe the thickness of the cutting edge portion **11** from 40 to 350 micrometers from the tip, with constants $c=0.2$ and $d=1.5$.

(51) According to another aspect, the thickness of the cutting edge portion **11** of the blade may have the following thickness configuration as detailed in following Table 5.

(52) TABLE-US-00005 TABLE 5 Blade profile parameters according another aspect

Distance from the tip	Thickness [μm]
5	1.82
20	5.82
30	8.33
40	10.84
50	13.35
100	25.90
150	38.45
200	47.38
250	56.25
300	65.13
350	74.00

(53) Further, the thickness of the blade profile may be described by the above mentioned mathematical formulas (A) and (B).

(54) Accordingly, formula (A) may describe the thickness of the cutting edge portion **11** from 0 to 20 micrometers, with constants $a=0.47$ and $b=0.84$. Formula (B) may describe the thickness of the cutting edge portion **11** from 20 to 150 micrometers, with constants $c=0.251$ and $d=0.800$. Formula (B) may also describe the thickness of the cutting edge portion **11** from 150 to 350 micrometers, with constants $c=0.1775$ and $d=11.8750$.

(55) According to yet another aspect, the thickness of the cutting edge portion **11** of the blade may have the following thickness configuration as detailed in the following Table 6.

(56) TABLE-US-00006 TABLE 6 Blade profile parameters according to yet another aspect

Distance from the tip	Thickness [μm]
5	1.60
20	4.80
30	7.00
40	9.15
50	11.25
100	22.44
150	31.26
200	40.86
250	50.28
300	59.57
350	68.75

(57) Further, the thickness of the blade profile may be described by the above mentioned mathematical formula (A).

(58) Accordingly, formula (A) may describe the thickness of the cutting edge portion **11** from 0 to 20 micrometers, with constants $a=0.45$ and $b=0.79$. Formula (A) may also describe the thickness of the cutting edge portion **11** from 20 to 350 micrometers, with constants $a=0.296$ and $b=0.93$.

(59) According to a further aspect, the thickness of the cutting edge portion **11** of the blade may have the following thickness configuration, as detailed in the following Table 7.

(60) TABLE-US-00007 TABLE 7 Blade profile parameters according to a further aspect

Distance from the tip	Thickness [μm]
5	1.96
20	5.93
30	8.54
40	11.06
50	13.52
100	25.24
150	36.35
200	47.10
250	56.10
300	65.10
350	74.10

(61) The thickness of the blade profile may be described by the above mentioned mathematical formulas (A) and (B).

(62) Accordingly, formula (A) may describe the thickness of the cutting edge portion **11** from 0 to 20 micrometers, with constants $a=0.54$ and $b=0.80$. Formula (A) may also describe the thickness of the cutting edge portion **11** from 20 to 200 micrometers, with constants $a=0.40$ and $b=0.90$. Formula (B) may describe the thickness of the cutting edge portion **11** from 200 to 350 micrometers, with constants $c=0.18$ and $d=11.10$.

(63) All the above aspects, which relate to the tip **14** and to the cutting edge portion **11** of the razor may be described by formula (A) and formula (B) or with the combination of both formulas. The formulas (A) and (B) may describe different sections measured from the tip **14** of the razor.

(64) The razor blade substrate **10** including the cutting edge portion **11** may be made of stainless steel. A suitable stainless steel may include in weight 0.62-0.75% of carbon, 12.7-13.7% of chromium, 0.45-0.75% of manganese, 0.20-0.50% of Silicon, No more than traces of Molybdenum, Balanced iron. Balanced iron can mean that iron comprises the rest of the material

by weight.

(65) Other stainless steels may also be used. Other materials which are known as razor blade substrate materials, may also be considered.

(66) The following manufacturing steps of a razor blade may be described below.

(67) The blade substrate **10** including a cutting edge portion **11** having a profiled geometry and having a tapering geometry with two substrate sides **12**, **13** converging toward a substrate tip **14**, may be covered by a strengthening coating **16** deposited on the razor blade substrate at least at the cutting edge portion **11**. Coating layers may be implemented on the blade edge substrate to improve the hardness of the blade edge and to thereby enhance the quality of the shaving.

(68) The coating layers may reduce wear of the blade edge, may improve the overall cutting properties and may prolong the usability of the razor blade.

(69) The strengthening coating **16** covering the substrate tip **14**, may have a profiled geometry and may have a tapering geometry with two coating sides converging toward a coating tip. As shown in FIG. 3, the blade edge substrate **10** may be coated with a strengthening coating layer **16** and a lubricating layer **17**. The lubricating layer **17**, which may comprise fluoropolymer, is commonly used in the field of razor blades for reducing friction during shaving. The strengthening coating layer **16** may also be used for mechanical properties. For example, the strengthening coating layer **16** may include titanium and boron. More precisely, the strengthening coating layer **16** may be made of titanium and boron with a low content of impurities. The content of impurities may be kept as low as economically possible. The strengthening coating layer **16** may be prepared with various proportions of titanium and boron within the layer. Other aspects may include a mixture of chromium and carbon, DLC, amorphous diamond, or other similar materials. The cutting edge portion **11** of the blade may be covered by an interlayer **15**. For example, the interlayer **15** may be made of Titanium, notably in the case of a titanium- and boron-containing strengthening coating layer **16**. In a case where the blade may be covered by a Titanium interlayer **15**, the interlayer **15** may be implemented prior to the strengthening coating layer **16**. Thus, the coating layer configuration of the cutting edge portion **11** of the blade may include a Titanium interlayer **15** covering the cutting edge portion **11** of the blade and strengthening coating layer **16** covering the Titanium interlayer **15**. Further, the strengthening coating layer **16** may be covered by a top layer **20**. For example, a top layer may be made of Chromium. The top layer **20** including Chromium may also be covered by a lubricating layer **17**, which may include a fluoropolymer, as shown on FIG. 4.

(70) The blade may be fixed or mechanically assembled to a razor head, and the razor head itself may be part of a razor. The blade may be movably mounted in a razor head, and mounted on springs which urge it toward a rest position. The blade may be fixed, notably welded to a support **29**, notably a metal support with a L-shaped cross-section, as shown in FIG. 8a. Alternatively, the blade may be an integrally bent blade, as shown on FIG. 8b, where the above disclosed geometry applies between the blade tip and the bent portion **30**.

Claims

1. A razor blade comprising a substrate including a cutting edge having a symmetrical tapering geometry with two substrate sides and a planar portion ending in a sharpened tip, the tapering geometry having a thickness of between 1.55 and 1.97 micrometers measured at a distance of five micrometers from the tip, to a thickness of between 4.60 and 6.34 micrometers measured at a distance of twenty micrometers from the tip, to a thickness of between 19.8 and 27.12 micrometers measured at a distance of hundred micrometers from the tip, wherein the thickness of the cutting edge of the substrate is described with the following mathematical formulas:

$$t=a.\text{Math.}(x.\text{sup.}b) \quad (\text{A})$$

$$t=(c.\text{Math.}x)+d \quad (\text{B}) \text{ wherein, } a \text{ and } c \text{ are constants from an interval } (0, 1), b \text{ is a constant from}$$

an interval (0.5, 1), d is a constant from an interval (0.5, 20), x refers to a distance from the tip in micrometers and t refers to the thickness of the blade in micrometers, and wherein equation (A) is used to determine the thickness of the cutting edge from the tip to a transition point, and equation (B) is used to determine the thickness of the cutting edge starting from the transition point, wherein a blade thickness increase rate is defined by a slope from the sharpened tip to the transition point, wherein the blade thickness increase rate in a region between the tip to the transition point is greater than the blade thickness increase rate of the cutting edge starting from the transition point.

2. The razor blade according to claim 1, wherein the tapering geometry of the substrate further includes a thickness of between 6.50 and 8.94 micrometers measured at a distance of thirty micrometers from the tip, wherein the tapering geometry of the substrate further includes a thickness of between 8.40 and 11.54 micrometers measured at a distance of forty micrometers from the tip, wherein the tapering geometry of the substrate further includes a thickness of between 10.30 and 14.13 micrometers measured at a distance of fifty micrometers from the tip, wherein the tapering geometry of the substrate further includes a thickness of between 29.30 and 40.11 micrometers measured at a distance of hundred fifty micrometers from the tip, wherein the tapering geometry of the substrate further includes a thickness of between 38.80 and 49.74 micrometers measured at a distance of two hundred micrometers from the tip.
3. The razor blade according to claim 1, wherein the tapering geometry of the substrate further includes a thickness of between 48.30 and 59.37 micrometers measured at a distance of two hundred fifty micrometers from the tip.
4. The razor blade according to claim 1, wherein the tapering geometry of the substrate further includes a thickness of between 57.80 and 69.00 micrometers measured at a distance of three hundred micrometers from the tip.
5. The razor blade according to claim 1, wherein the tapering geometry of the substrate further includes a thickness of between 67.30 and 78.62 micrometers measured at a distance of three hundred fifty micrometers from the tip.
6. The razor blade according to claim 1, wherein the substrate is a stainless steel comprising, in weight: 0.62-0.75% of carbon, 12.7-13.7% of chromium, 0.45-0.75% of manganese, 0.20-0.50% of Silicon, no more than traces of Molybdenum, and balanced iron.
7. The razor blade according to claim 1, wherein the substrate is covered by a strengthening coating.
8. The razor blade according to claim 7, wherein the strengthening coating comprises Titanium and Boron.
9. The razor blade according to claim 8, wherein the substrate is covered by an interlayer and the interlayer is covered by the strengthening layer.
10. The razor blade according to claim 8, wherein the strengthening layer is covered by a top layer.
11. The razor blade according to claim 10, wherein the top layer is covered by a polytetrafluoroethylene (PTFE) layer.
12. The razor blade according to claim 1, wherein the blade thickness increase rate in a region between a distance from 40 micrometers to 300 micrometers from the tip is less than the blade thickness increase rate from the tip of the blade to the distance of 40 micrometers from the tip.
13. The razor blade according to claim 12, wherein the blade thickness increase rate in a region between a distance from 40 micrometers to 350 micrometers from the tip is less than the blade thickness increase rate from the tip of the blade to the distance of 40 micrometers from the tip.
14. The razor blade according to claim 1, wherein the transition point is between about 20 micrometers from the tip to about 40 micrometers from the tip.
15. The razor blade according to claim 1, wherein equation (A) is used to determine the thickness of the cutting edge from the tip to about 40 micrometers from the tip, where $a=0.5$ and $b=0.8$, and either equation (A) or equation (B) with constants $c=0.2$ and $d=1.5$ used to determine the thickness of the cutting edge from about 40 micrometers from the tip.

16. The razor blade according to claim 1, wherein equation (A) is used to determine the thickness of the cutting edge from the tip to about 20 micrometers from the tip, where $a=0.47$ and $b=0.84$, and either equation (A) or equation (B) with constants $c=0.251$ and $d=0.800$ used to determine the thickness of the cutting edge from about 20 micrometers from the tip.

17. The razor blade according to claim 1, wherein equation (B) is used to determine the thickness of the cutting edge from about 150 micrometers to about 350 micrometers, with constants $c=0.1775$ and $d=11.8750$.

18. The razor blade according to claim 1, wherein equation (A) is used to determine the thickness of the cutting edge from the tip to about 20 micrometers from the tip, where $a=0.45$ and $b=0.79$, and either equation (A) or equation (B) with constants $c=0.296$ and $d=0.93$ used to determine the thickness of the cutting edge from about 20 micrometers from the tip.

19. The razor blade according to claim 1, wherein equation (A) is used to determine the thickness of the cutting edge from the tip to about 20 micrometers from the tip, where $a=0.54$ and $b=0.80$, and equation (A) with constants $a=0.40$ and $b=0.90$ used to determine the thickness from about 20 micrometers from the tip to about 200 micrometers from the tip, and equation (B) with constants $c=0.18$ and $d=11.10$ is used to determine the thickness from about 200 micrometers to about 350 micrometers from the tip.
